Finite Symbolic Mechanics: The Mathematics of Discrete Dynamical Systems

Discrete Dynamical Systems through the Lens of Geofinitism

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Symbolic Beginnings

All mathematics begins as sequences of symbols arranged upon a page. These symbols are processed serially, step by step, by the reader. Clocks and instruments measure the passage of those steps, and further symbols record the assigned times. This serial, symbolic, and measured framework constitutes the universal substrate of mathematical thought.

Within this substrate, over millennia, communities of mathematicians have stabilised distinct collections of tools, conventions, and modes of interpretation. These collections acquired names — arithmetic, geometry, analysis, algebra, number theory, topology — and grew into recognised fields. Each field functions as a lens: a way of selecting, arranging, and interpreting symbols according to its own foundational commitments.

A problem can be considered through any of these lenses. Each lens traverses a different subset or transformation of the symbolic sequence and reveals a different picture. The story of mathematics is therefore the story of lens construction and lens application. Creating a new lens is itself a significant dynamical process — one that is unfolding in the present work.

The Building of Lenses

The history of mathematics records repeated acts of lens construction. Early geometry stabilised a lens of straightedge and compass. The calculus introduced continuous change and limits. Complex numbers extended algebra beyond the real line. Non-Euclidean geometries arose when the parallel postulate was recognised as a commitment rather than a necessity. In each case, explicit adjustment of foundational commitments opened previously illegible structure to view.

The Lens of Nonlinear Dynamics

In the twentieth century the lens of nonlinear dynamics brought new clarity to systems governed by change. Edward Lorenz's work on convection revealed deterministic rules producing bounded yet aperiodic motion and geometric attractors of striking complexity. Takens' delay embedding theorem provided a method to reconstruct attractor geometry from scalar observations, provided the system satisfied assumptions of smoothness, underlying

manifolds, and infinite precision. This lens illuminated phenomena across physics and engineering but left discrete symbolic processes — integer iterations and computational rules — largely outside its clear domain.

Geofinitism: A New Lens and Its Foundational Commitments

Geofinitism extends the dynamical lens into the domain of finite symbolic systems. It recognises that all mathematical work occurs through finite sequences of symbols processed with bounded precision. Creating this lens is a dynamical act of mathematical evolution: a deliberate construction of new commitments suited to the symbolic and measured character of discrete systems.

Every mathematical lens is defined by its foundational commitments — explicit statements about the nature of objects, measurement, representation, and reality. These commitments constitute the separatrix between lenses: when a system violates the commitments of one lens, analysis becomes incommensurate and results shift from productive idealisation to category error. The commitments of Geofinitism, which arise from its developing philosophy, may be articulated in five pillars.

Commitment 1: Geometric Container Space
All knowable structure emerges as trajectories and positions within a finite-dimensional geometric manifold. Meaning is not abstract retrieval but navigation through this space; agreement is convergence of trajectories, misunderstanding is divergence. *(Geofinitism Pillar 1)*

Commitment 2: Approximations and Measurements
All symbols and representations are finite transductions — lossy compressions of exogenous and endogenous reality. Every measurement introduces distortion and uncertainty; perfect correspondence is impossible, and only useful approximation remains. *(Geofinitism Pillar 2)*

Commitment 3: Dynamic Flow
Meaning and structure are inherently dynamical. Symbols and concepts flow through cascading layers, forming attractors at every scale. What appears static is merely slow-moving geometry; memory is the tendency to return to established trajectories. *(Geofinitism Pillar 3)*

Commitment 4: Useful Fiction
Classical infinite-object mathematics and idealised constructs function as useful fictions rather than literal descriptions of reality. Their validity rests on utility within regimes where their assumptions approximately hold, grounded in measurement rather than any claim of direct correspondence to an external or Platonic truth. Classical mathematical proofs themselves live as documents within existing attractor basins of prior commitments. *(Geofinitism Pillar 4)*

Commitment 5: Finite Reality
All measurements and representations are fundamentally bounded. There are no actual infinities or perfect zeros in the knowable world — only finite approximations with inherent uncertainty. Unlike earlier systems of commitments, Geofinitism holds uncertainty as a first-class principle and can therefore treat itself as an approximation within its own framework. *(Geofinitism Pillar 5)*

> These pillars are not presented as final dogma but as the current, coherent articulation of a living philosophy still under development. They distinguish Geofinitism sharply from frameworks that treat continuity, infinite precision, or Platonic objects as primary.

Geofinitism articulates the philosophical commitments of the lens. **Finite Symbolic Mathematics** is its practical implementation—the application of these commitments to the study of finite symbolic processes under bounded measurement.

Revealing Hidden Structure: The Collatz Example

These five commitments do more than articulate a philosophy; they re-license techniques that classical commitments forbid. Takens' delay embedding theorem, the standard lens for attractor reconstruction, assumes a smooth diffeomorphism, continuous dynamics, and infinite precision—conditions the Collatz map manifestly violates. Under classical commitments, applying delay embedding to Collatz is formally unjustified. Geofinitism replaces those commitments with its own pillars, creating a framework in which the same reconstruction becomes legitimate. The Finite-Symbol Embedding Theorem, developed in the companion paper, formalises this shift: it provides the licensing conditions under which discrete, finite-precision systems can be treated geometrically without violating the lens's own rules. With that license in

hand, we can now examine what becomes visible when the Geofinite lens is applied to the Collatz problem.

The Collatz iteration offers an ideal test case. Defined by a straightforward rule—given any positive integer n, if n is even divide by two; if n is odd multiply by three and add one—the process is then repeated indefinitely. The conjecture asserts that every such sequence eventually reaches the cycle $1 \rightarrow 4 \rightarrow 2$. Despite its elementary character and extensive computational verification, the conjecture has resisted formal proof for more than eighty years.

The Riemann Hypothesis presents a parallel case, examined in full elsewhere. There, the same Geofinite commitments reveal that the critical line $\mathrm{Re}(s) = 1/2$ corresponds to the geometric centre of base-10 computational space—an attractor arising from the alphonic structure of the symbolic substrate rather than a Platonic coincidence. Both examples follow the same pattern: classical lenses reveal emergent structures while their own commitments forbid the analysis that would explain them; Geofinitism provides the licensing framework that makes the structure legible.

When Collatz sequences are reconstructed via delay coordinates under the Geofinite commitments, a coherent geometric object appears: a comma-shaped manifold through which all examined trajectories pass before converging toward the (1, 4, 2) cycle neighbourhood. Multiple independent analyses converge on the same picture—bounded chaotic motion, high determinism, low-dimensional scaling, and a single connected basin. The structure was always latent in the symbolic sequence. The Geofinite lens makes it visible.

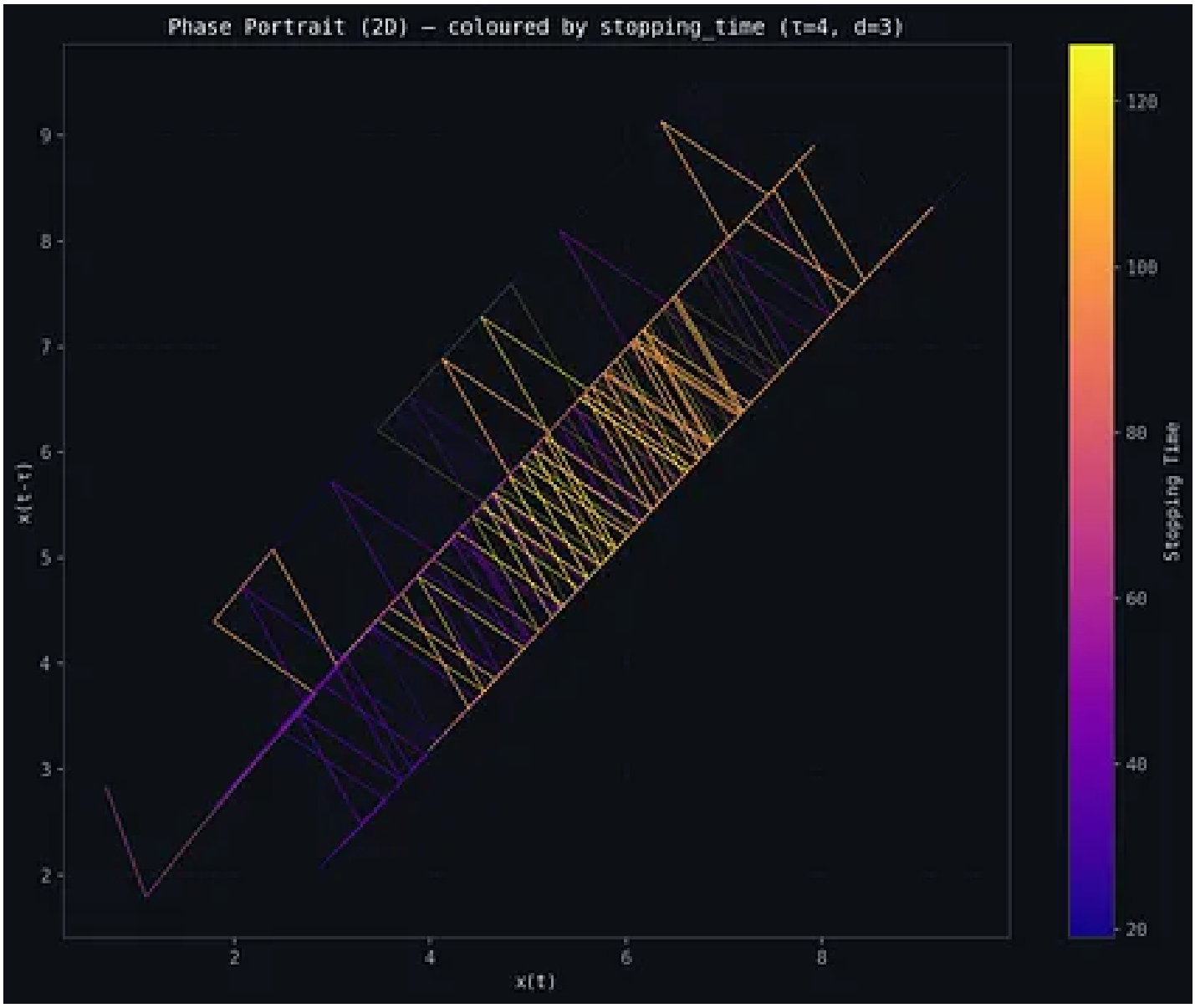

Figure 1: Phase portrait of Collatz trajectories in reconstructed delay-embedded space.

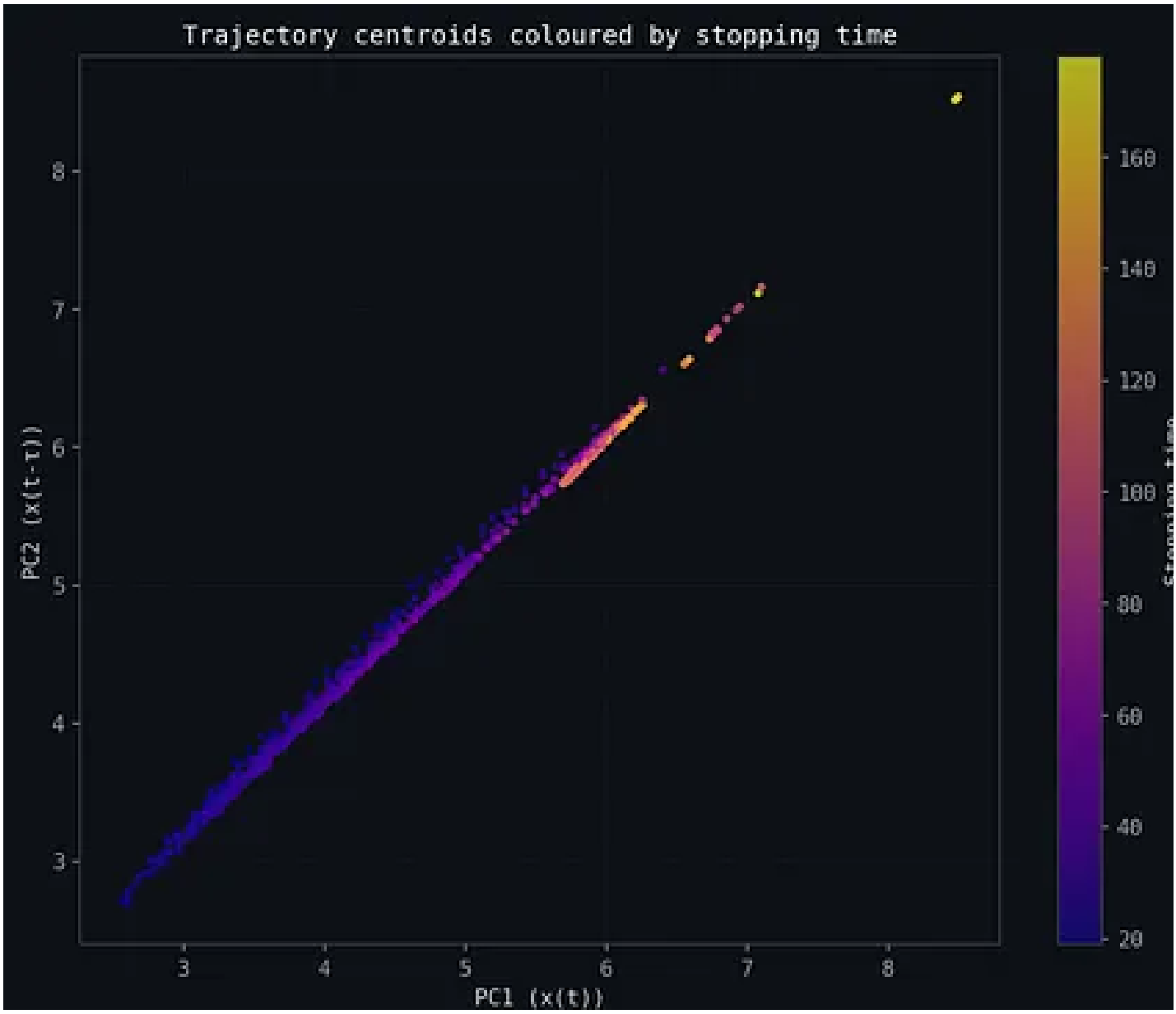

Figure 2: Clustering of trajectory centroids demonstrating a single connected attractor basin.

Implications and the Dynamical Process of Lens Creation

The creation of Geofinitism demonstrates that building a new lens is itself a dynamical process within mathematics. By making commitments explicit and aligning them with the finite, symbolic, measured nature of discrete systems, previously resistant problems gain geometric legibility. The approach suggests similar reframing for other iterative maps, cellular automata, and computational processes.

Yet articulation alone does not constitute establishment. A lens becomes recognised only when it achieves consensus—when a community of practitioners adopts its commitments, replicates its results, and stabilises its methods as a shared attractor basin. This process is itself dynamical, governed by recognition, replication, and institutional stabilisation rather than by logical necessity alone. Geofinitism is offered in that spirit: not as a finished edifice demanding immediate acceptance, but as a coherent framework whose utility can be tested through application to problems—like Collatz and Riemann—that have resisted resolution under classical commitments.

The legitimacy gap that accompanies any new lens is not a flaw in the lens but a feature of the dynamical landscape. Established lenses derive their authority from consensus, which functions as an attractor: a stable configuration of practitioners, journals, funding, and institutional recognition. When one works within such a lens, claims are automatically licensed by the community's prior commitments. A new lens lacks this consensus by definition. Its proposals are not heard as modifications from within but as perturbations from without, and the natural response of any stable system is to deflect inputs that would shift its attractor configuration. This resistance is not obstruction; it is the dynamical condition that prevents lenses from changing too easily and thereby preserves the stability that enables cumulative progress.

The position of the lens-builder, then, is inherently that of an outsider relative to the established basins. This is not incidental but structural. The trajectory of a new lens from proposal to consensus follows identifiable dynamical patterns: first articulation, then demonstration, then replication by others, and finally stabilisation as a new attractor. The present work occupies the early phases of this process. Whether Geofinitism becomes an established lens depends on whether others find its geometry legible, its re-licensing compelling, and its results replicable across the discrete systems it claims to illuminate.

The companion papers develop this programme fully: the philosophical examination of lenses and Geofinitism, the formal Finite-Symbol Embedding Theorem that licenses reconstruction under finite-precision conditions, and the empirical demonstration on the Collatz process. Further applications—including the alphonic resolution of the Riemann Hypothesis—extend the framework to other domains where classical commitments have obscured rather than illuminated.

Mathematics advances through many routes. One of the most consequential is the deliberate construction and application of new lenses whose commitments match the systems under study.

The Collatz example shows that sometimes the hidden structure requires only the right lens to become visible. **Finite Symbolic Mathematics**—the practical implementation of Geofinitism—offers such a lens. By treating mathematics as the study of finite symbolic processes under bounded measurement, it re-licenses techniques that classical commitments forbid and reveals structure that classical lenses obscure. Whether it achieves consensus as a foundational field is a question that lies not in the lens alone but in the dynamical process of its reception.licenses reconstruction under finite-precision conditions, and the empirical demonstration on the Collatz process. Further applications—including the alphonic resolution of the Riemann Hypothesis—extend the framework to other domains where classical commitments have obscured rather than illuminated.

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The Tryptic of Finite Symbolic Mechanics Papers:

- Mathematics as Lenses: Geofinitism and the Reconstruction of Discrete Dynamical Structure
- The Finite-Symbol Embedding Theorem: Phase Space Reconstruction for Finite Symbolic Dynamical Systems
- A Nonlinear Dynamical Reconstruction of the Collatz Process via Delay Embedding

Further Essays based on the concepts of Geofinitism and Finite Symbolic Mathematics can be found here with the wider Attralucian Essays:

- Geometry in Geofinitism — A Thesis on Geofinitism and the Geometry.
- Base Invariance Dissolution - I — The Geofinite Dissolution of the Invariant
- Base Invariance Dissolution - II — The Alphonic Proofs
- The Generon — The Process of Symbolic Instantiation
- A New Face of Pi — Pi as Never Seen Before
- Is This A Philosophical Essay — A Measurement For You to Make
- The Ket Limit — Where Geofinitism Meets Quantum Mechanics
- Resolving the Riemann Hypothesis — The Dynamics of Mathematics
- Resolving the Fermi Paradox — Why We Can't Hear the Aliens
- Why Arithmetic Works — Beyond Russell's 2+2=4, a Geofinite Addition
- Verbs and Nouns — The Attractor and the Choice
- Geometric Numbers - I — Takens: Replacing Complex Numbers
- Geometric Numbers - II — Takens: Foundations for Analytic Functions

And for more on Geofinitism:

The Attralician Essays
Foundation Papers
Geofinitism Web Site

> *Omne quod est, finitum est; tantum per mensuram cognosci potest*
> *Everything that exists is finite; it can only be known by measure*
>
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